

Thermomechanical behavior of steel structural configurations subjected to cyclic cryogenic conditions in Martian environment

Jing Li^a, Yongtao Bai^{a,*}, Sifeng Bi^b, Michael Beer^{c,d,e}

^a Department of Civil Engineering, Chongqing University, Chongqing, China

^b Department of Aeronautics and Astronautics Engineering, University of Southampton, Southampton, United Kingdom

^c Institute for Risk and Reliability, Leibniz Universität Hannover, Callinstr. 34, Hannover 30167, Germany

^d Department of Civil and Environmental Engineering, University of Liverpool, Liverpool L69 3GH, United Kingdom

^e International Joint Research Center for Resilient Infrastructure & International Joint Research Center for Engineering Reliability and Stochastic Mechanics, Tongji University, Shanghai 200092, China

Received 16 April 2025; received in revised form 14 June 2025; accepted 25 June 2025

Available online 30 June 2025

Abstract

Mars is a potential destination for future human colonization, and the construction of sustainable habitats capable of withstanding its extreme low-temperature fluctuations is critical. Numerical simulations are employed to assess the thermal and mechanical performance of three common extraterrestrial structural configurations — Arch, Dome, and Cylinder — under Martian diurnal temperature fluctuations. The study employs stainless steel as the primary material. Key focus areas include stress concentration, displacement, and fatigue failure in the structures. The key findings are: (1) A 3-meter-thick regolith shielding significantly slows heat conduction, mitigating the impact of extreme temperature fluctuations on the steel structure. A fitted curve demonstrates how the structure gradually reaches thermal equilibrium under sustained temperature cycling, correlating temperature cycles with the minimum surface temperature of the steel structure; (2) The introduction of thermal cycles and internal-external pressure differences leads to changes in structural performance, with the regolith shielding providing the most effective protection for the Cylinder configuration; (3) Expansion and contraction caused by thermal cycles are constrained by the fixed base plate, resulting in structural failure originating from the bottom connections. Comparative analysis reveals that the Dome configuration offers superior load distribution and spatial efficiency, making it optimal for Martian habitats subjected to uniform pressure. The results highlight the thermal responses and fatigue behaviors of different steel structural configurations under Mars's extreme environmental conditions, providing scientific support for the selection and design of Martian habitat structures, offering strategies for optimization in structural design, and contributing to the development of reliable structural solutions for future Mars exploration and human settlement.

© 2025 Published by Elsevier B.V. on behalf of COSPAR.

Keywords: Martian habitat; Structural configuration; Thermal cycling; Numerical simulation; Thermal fatigue performance

1. Introduction

Mars is recognized as the most promising candidate for human colonization due to its Earth-like diurnal cycle and

seasonal variations (Scheer et al., 2007). With the Mars colonization plan proposed by SpaceX, utilizing the Starship Super Heavy launch system to provide the technological foundation for large-scale personnel and cargo transport, the construction of habitats that can withstand the extreme Martian environment is a critical engineering challenge. The extreme environmental conditions on the Martian surface — characterized by severe diurnal temperature fluctuations ($\Delta T > 100^\circ\text{C}$), low atmospheric pressure ($\sim 0.6\text{ kPa}$),

* Corresponding author.

E-mail addresses: lijing2023@stu.cqu.edu.cn (J. Li), bai.yongtao@cqu.edu.cn (Y. Bai), Sifeng.Bi@soton.ac.uk (S. Bi), beer@irz.uni-hannover.de (M. Beer).

and intense cosmic radiation (Petrov and Ochsendorf, 2005; Naser, 2019b; Katzer et al., 2021; Benaroya, 2017; Petrov, 2004) — impose dual requirements on both material selection and structural configuration for habitats. Notably, the first-ever successful application of stainless steel in Starship's hull validates its engineering feasibility in deep-space extreme environments: the material exhibits enhanced strength at cryogenic temperatures, combined with superior weldability, cost-effective manufacturability, and modular construction advantages, providing a technologically transferable material solution for Martian habitat development.

Following the establishment of fundamental material systems, the exploration and optimization of geometric configurations for extraterrestrial habitat structures becomes a critical phase for ensuring the safety of long-duration missions. While structural design and analysis studies are available for a range of possible structural configurations (Task Committee on Lunar Base Structures, 1992; Sherwood and Touns, 1992; Malla et al., 1995; Ruess and Benaroya, 2006; Tripathi and Malla, 2023), research addressing their thermomechanical performance under operational constraints remains limited. Mottaghi and Benaroya (2015) proposed an igloo-shaped magnesium structure based on sintered regolith and performed thermal analysis using finite element software, investigating the impact of regolith shielding. Tripathi et al. (2023) introduced a method to determine the surface temperature and temperature distribution within the wall thickness of a hemispherical dome lunar habitat structure constructed from lunar regolith simulants. Malla et al. have also contributed significantly to this field; in 1995, they analyzed the structure under static loading conditions using finite element methods. In 2011, they employed ABAQUS to perform a static analysis of an aluminum frame + Kevlar members structure, considering the weight of a 1-meter-thick regolith shielding layer applied externally to the structure, along with internal pressurization loads acting perpendicular to the membrane pressure boundary (Malla and Filburn, 2011). Subsequently, ABAQUS was used to simulate the impact of internal membranes, temperature variation, and different internal pressures on the frequency response of a rectangular lunar habitat concept (Malla and Brown, 2012). Existing research, however, lacks in-depth configuration studies and performance evaluations specifically based on the actual conditions of the Martian environment. In particular, the specific effects of extreme temperature fluctuations on structural performance remain inadequately studied. The complex thermodynamic challenges faced by structures during long-term use on Mars have yet to be fully assessed, and efficient, stable structural configuration selection remains inadequately validated.

Given the limitations of terrestrial experimental facilities in fully replicating Martian environmental conditions, numerical simulation offers significant advantages in investigating environmental coupling effects, making it a critical methodology for exploring habitat structural design

(Zhang and Li, 2022). This study employed numerical simulation to conduct a comparative analysis of the thermal response and fatigue behavior of three typical structural configurations — Arch, Dome, and Cylinder — under Martian cryogenic thermal cycling, with a focused investigation of the stress concentration, displacement deformation, and crack propagation locations and paths of the structures. The goal is to provide theoretical support for the selection and optimization of future Martian habitat configurations, offer a reference for developing more reliable Martian construction design standards, and contribute to the sustainable development of Martian habitats.

2. Load combinations and structural configurations

2.1. Load combinations

Martian structures face vastly different environmental conditions, external loads, and material properties compared to terrestrial structures. Even minor damage, such as a small crack causing air leakage, can lead to freezing, expansion, crack propagation, and eventual collapse. Therefore, unlike Earth structures, Martian structural design must not tolerate any form of damage and should be designed considering the most adverse load combinations (Kuster and Delorit, 2023).

2.1.1. Stress loads

Understanding the mechanisms and cumulative effects of various loads in the Martian environment is crucial (Soureshjani et al., 2023). The most common load combination includes gravitational forces and high internal pressures. This section analyzes stress loads from the following aspects (Table 1):

- (1) Gravity load: With a gravitational acceleration of 3.73 m/s^2 on Mars (NASA, 2004), structures experience lighter gravitational forces than on Earth.
- (2) External atmospheric pressure: Mars's atmosphere is thin, with an average atmospheric pressure of about 1 % of Earth's (Williams, 2004), which varies by location and season (NASA reports a surface pressure of approximately 0.61 kPa, while the Viking 1 lander observed values between 0.69 kPa and 0.9 kPa (Bucklin et al., 2004). Although relatively small compared to the high internal pressures, it must be considered a distinct load before the structural components are fully installed.
- (3) Internal pressure: To sustain Earth-like living conditions, Martian habitats must withstand internal pressures of 101.325 kPa, especially at joints and seals.
- (4) Regolith shielding: It is a critical design feature for Martian habitats, offering protection against tensile stresses, radiation, micrometeorite impacts, and extreme temperature fluctuations. Studies indicate that approximately 3 m of regolith can effectively mitigate these external hazards, making it an essential

Table 1
Load combinations for Martian habitat structures.

Load Type	Application Position	Value	Notes
Gravitational Load	structure	$D_M = mg_M$	$g_M = 3.73 \text{ m/s}^2$
Internal Pressure	Internal	101.325 kPa	To maintain suitable living conditions for humans
Atmospheric Pressure	External	900 Pa	Martian surface atmospheric pressure
Regolith shielding	External	$=\rho_{RS} \cdot g_M \cdot h_{RS}$	3 m thickness
Other Loads	External	≈ 0	Compared to the high internal pressure, it is small enough to be considered negligible.
Thermal stress	Structure	T_M	The effect of thermal influence is based on heat transfer coefficients (α), elastic modulus (E), temperature variation (ΔT), and several other structural calculations (Callister and Rethwisch, 2020). Further discussion will be provided in section 2.1.2.

Note: g_M – Gravitational acceleration on Mars; D_M – Gravitational load of the structure; ρ_{RS} – Density of regolith shielding; h_{RS} – Thickness of regolith shielding; T_M – Thermal stress on Mars.

protective measure for Martian structures (Mottaghi and Benaroya, 2015).

- (5) Other loads: Additional factors such as dust accumulation, thermal stress, wind, marsquakes, and meteoroid impacts may also affect the structure, but these are generally minor and can often be neglected, especially in comparison to high internal pressures. For instance, wind loads less than 10 Pa (NASA, 2015) can be considered negligible; whereas, special loads like meteor and meteoroid impacts are rare and are not discussed here as routine loads.

2.1.2. Thermal stress

The Martian atmosphere is thin and lacks an effective greenhouse effect, resulting in extremely low and fluctuating surface temperatures. This extreme thermal cycling poses challenges to the stability and durability of habitat structures (Kuster and Delorit, 2023). Thermal stress arises from rapid temperature changes: as materials expand or contract, constraints within the structure may lead to thermal fatigue cracks.

Mars’s rotational period is nearly identical to that of Earth, with an average Martian day lasting 24 h, 39 min, and 35 s. Its axial tilt of 25.19° is similar to Earth’s

23.44°, resulting in four seasons, each lasting about twice as long as Earth’s (<https://starwalk.space/zh-Hans/news/mars-the-ultimate-guide>). Using data from NASA’s InSight Mission and the diurnal temperature variation on Mars (Toklu, 2005; <https://mars.nasa.gov/insight/mision/weather/>), a regular temperature cycling curve (Fig. 1) was constructed to model surface temperature fluctuations. The cycle assumes a 24-hour period with 16 h of cooling and 8 h of warming, ranging from −120 °C to +20 °C. This simplified curve represents the most extreme conditions on Mars, and its application as a boundary condition allows for the quantification and evaluation of temperature variations effects on structural performance.

2.2. Structural configurations selection

Unlike terrestrial building design, the unique conditions of the Martian environment require appropriate adjustments to traditional design methods, particularly due to low gravity and extreme temperature fluctuations. On Earth, gravitational loads dominate, making parabolic arches more common than semicircular arches, as they effectively minimize the accumulation of internal moments. However, on Mars, internal pressure outweighs

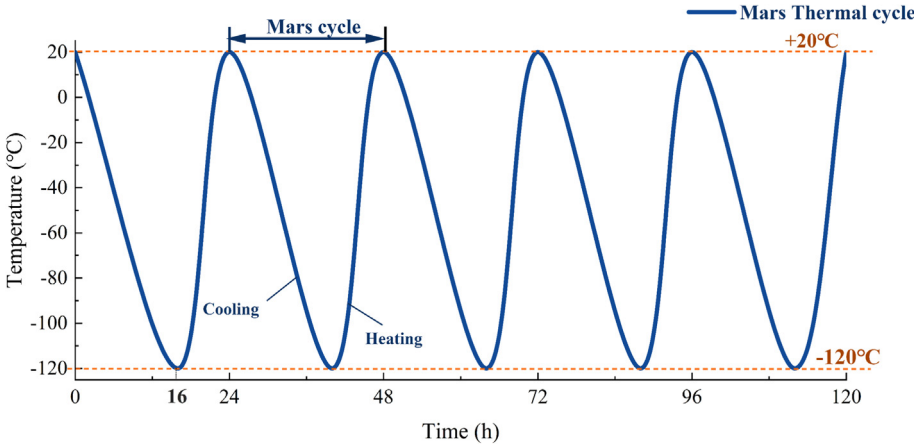


Fig. 1. Simulated temperature cycling curve of Mars.

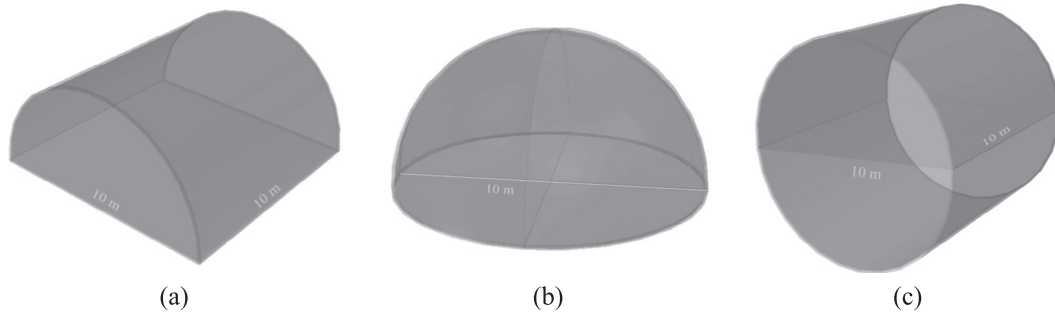


Fig. 2. Common geometric configurations for extraterrestrial structure design: (a) Arch; (b) Dome; (c) Cylinder.

gravitational loads, making semicircular arches more suitable for efficiently transferring loads to the ground with minimal moments (Ruess and Benaroya, 2006). Therefore, in Martian habitat design, structural configurations should prioritize more efficient and economical geometries, such as spherical, cylindrical, or domed shapes, which help minimize structural stress and reduce material usage (Toklu, 2005; Jarvstrat and Toklu, 2004).

Three commonly used and widely applied architectural configurations — semi-circular arch (hereinafter referred to as Arch), Dome, and Cylinder — are selected as the subjects (Fig. 2). These structural forms are applied to varying degrees in the design of Martian habitats, primarily due to several advantages. Their geometric shapes are simple, offering high internal space utilization and reducing the complexity of design and construction, making them suitable for Mars's limited resources and technological constraints. Additionally, all three configurations support modular design, allowing for flexible expansion and adjustment as needed, facilitating modular assembly, transportation, and rapid deployment. These configurations are also single-layered independent structures, with simple structural connections, making them ideal for the resource limitations and task requirements of the Martian environment.

In terms of size selection, typical structural segments with a diameter of 5 m and a depth of 10 m were chosen to meet basic habitation needs, ensuring adequate vertical height and living space. For material selection, steel plates with a thickness of 100 mm were used as the primary material. In performance evaluation, the focus was primarily on the service performance during two stages: the construction phase with bare components and the phase with the regolith covering.

3. Materials and methods

The thermal fatigue process is a transient heat transfer problem, and the finite element method (FEM) provides a foundation for analyzing temperature fields and thermal stresses, crucial for simulating crack propagation — a task difficult to fully replicate in experiments (Bhattacharya and Ellingwood, 1998; Liu et al. 2008). The governing equations include:

1. Transient heat conduction equation (energy conservation) (Eq. (1)):

$$\rho C_P \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + Q \quad (1)$$

2. Thermal-elastic stress equilibrium equation (inertia neglected) (Eq. (2)):

$$\nabla \cdot \sigma + \rho b = 0, \quad \sigma = C : (\varepsilon - \alpha(T - T_{ref})I) \quad (2)$$

where $\varepsilon = \frac{1}{2}(\nabla u + (\nabla u)^T)$ represents the total strain tensor, and u denotes the displacement field.

FEM analysis of thermal stress, strain, and temperature fields offers valuable insights into thermal fatigue crack propagation under fluctuating thermal boundary conditions. Calculations were performed using ABAQUS, employing sequential thermodynamic coupling in FEM thermal analysis. The weak-form discretization for the numerical solutions are presented in Eqs. (3) and (4):

$$\begin{aligned} \int_{\Omega} \mathbf{N}^T (\rho C_P \frac{\partial T}{\partial t} - \nabla \cdot (k \nabla T)) d\Omega \\ = \int_{\Omega} \mathbf{N}^T Q d\Omega \text{ (Thermal analysis)} \end{aligned} \quad (3)$$

$$\begin{aligned} \int_{\Omega} \mathbf{B}^T \sigma d\Omega = \int_{\partial\Omega} \mathbf{N}^T t d\Gamma \text{ (Mechanical analysis, } \mathbf{B} \\ \text{: strain – displacement matrix, } \mathbf{N} \\ \text{: shape functions)} \end{aligned} \quad (4)$$

First, a thermal analysis was performed to obtain the temperature field $T(\mathbf{x}, t)$ of the structure, which was then applied as a body load to solve for stress distribution via the thermo-mechanical constitutive relationship in Eq. (5):

$$\sigma = C(T) : (\varepsilon - \alpha(T - T_{ref})I) \quad (5)$$

Iterative computation yields the time-dependent stress field $\sigma(\mathbf{x}, t)$ (Fig. 3).

3.1. Material properties

This study employs 304L stainless steel as the primary structural material for its notable benefits: (1) Exceptional

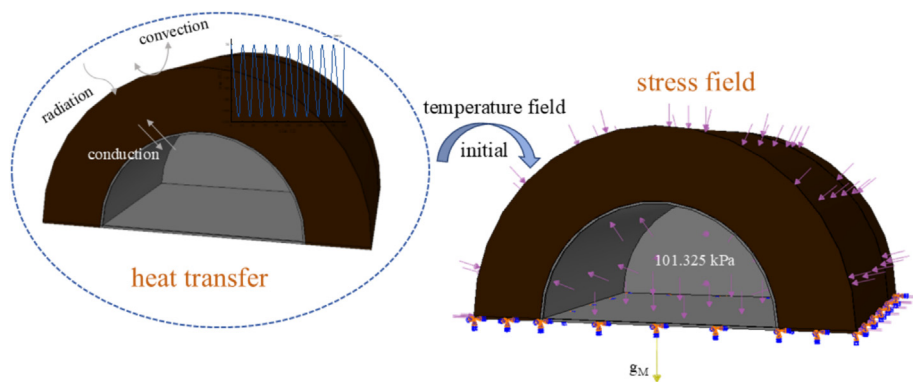


Fig. 3. Sequential thermodynamic coupling process in ABAQUS.

Table 2
The physical parameters of 304L steel (Kim, 1975).

Property	Unit	Value
Density	g/cm ³	$\rho = 7.9841 - 2.6506 \times 10^{-4}T - 1.1580 \times 10^{-7}T^2$
Specific heat	cal/g·K	$C = 0.1122 + 3.222 \times 10^{-5}T$
Thermal conductivity	W/cm·K	$K = 8.116 \times 10^{-2} + 1.618 \times 10^{-4}T$
Yield stress	MPa	205
Poisson's ratio	/	0.3
Young's modulus	GPa	20 °C: 193 −110 °C: 200 −120 °C: 202
Coefficient of linear thermal expansion	E ^{−5} /K	$B = 1.2632 \times 10^{-5} + 4.8544 \times 10^{-9}T + 7.2376 \times 10^{-13}T^2$

T – temperature in K.

low-temperature performance, maintaining superior fracture toughness even at −196 °C; (2) Cost-effectiveness, significantly lowering costs compared to common aerospace materials like aluminum and titanium alloys, and carbon fiber, while maintaining a minimum 40 % elongation rate; (3) Proven engineering reliability, evidenced by its use in SpaceX's Starship, which confirms its potential and reliability under extreme space conditions. These properties make 304L stainless steel well-suited to address the severe diurnal temperature fluctuations on Mars, providing a cost-effective solution for habitat construction. The specifications of 304L are detailed in Table 2.

Regolith, with its low thermal conductivity, effectively mitigates the impact of Martian surface temperature variations on structural thermal stress (Table 3).

Table 3
The physical parameters of Martian regolith.

Property	Unit	Value
Density	kg/m ³	1500 (Peters et al.,2008)
Radiation Shielding Thickness	m	3
Young's modulus	GPa	50
Poisson's ratio	/	0.3
Heat capacity	J/m ³ ·K	10 ⁶ (Spohn et al. 2024)
Thermal conductivity	W/m·K	0.036 (Spohn et al. 2024)

On the Martian surface, structures are mainly influenced by radiative and convective heat transfer, with key parameters listed in Table 4.

3.2. Boundary conditions and meshing

3.2.1. Initial and boundary conditions

To accurately simulate the temperature variations and thermal stresses of the structure in the Martian environment, appropriate boundary conditions must be set, and the interaction between structural components and regolith considered during meshing. Given extreme diurnal thermal variations and external environmental impacts on Mars, this study divides the service life into two distinct phases for analysis. In each stage, the structure is assumed to be in an intact initial state, ensuring a thorough evaluation of the temperature fatigue performance under both weathered and non-weathered conditions (Figs. 4 and 5).

- (1) Phase 1: Thermal fatigue performance during the construction phase

While regolith can be used to maintain a more consistent internal temperature of the structure, the structure itself must still endure cold temperatures during the installation phase (Jablonski and Ogden, 2008). Without regolith

Table 4
Structural material properties and model parameters.

Parameters	Unit	Magnitude
Stefan-Boltzmann's constant, σ	$\text{W}/(\text{m}^2 \cdot \text{K}^4)$	5.67×10^{-8} (Incropera et al. 1996)
Surface Emissivity, ε	/	0.9 (Mottaghi and Benaroya,2015)
Natural Convective Coefficient	$\text{W}/\text{m}^2 \cdot \text{K}$	10 (Mottaghi and Benaroya,2015; Emmel et al., 2007)

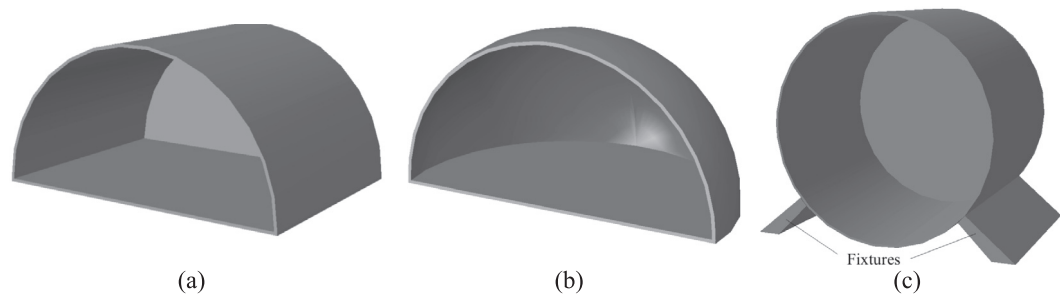


Fig. 4. Structural service phase 1 (without regolith coverage): (a) Arch-sectional view; (b) Dome-sectional view; (c) Cylinder-sectional view.

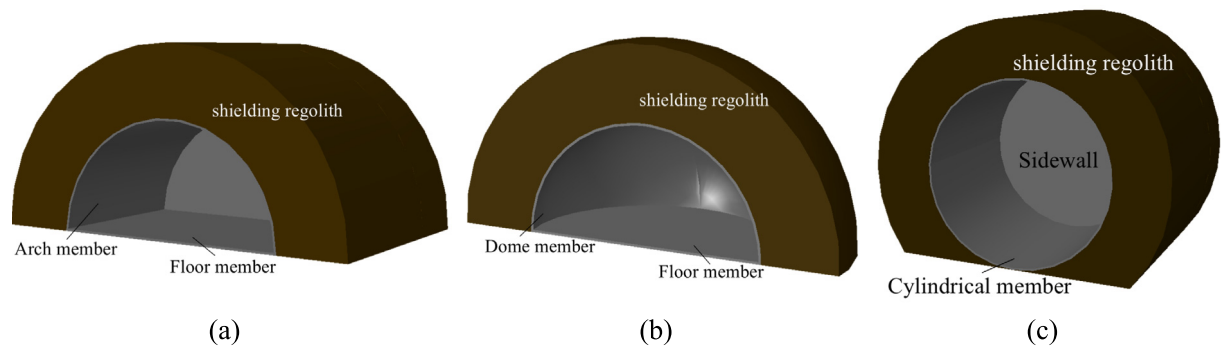


Fig. 5. Structural service phase 2 (with regolith coverage): (a) Arch-sectional view; (b) Dome-sectional view; (c) Cylinder-sectional view.

shielding, the structure is fully exposed to Martian environmental loads. This phase, crucial for the structure's lifecycle, assesses its performance under extreme conditions in the construction phase (Benaroya, 2017) (Table 5).

(2) Phase 2: Thermal Fatigue Performance of In-Service Structures

This phase analyzes the thermal fatigue performance and structural stability under long-term service conditions

with regolith protection. While the structure's surface is shielded by regolith that bears extreme Martian environmental loads, the controlled interior environment maintains stable habitat conditions approaching Earth's atmospheric pressure (Table 6).

3.2.2. Finite element mesh generation

The meshing quality of the finite element model directly impacts the accuracy of the analysis. To accurately capture

Table 5
Initial conditions for the simulation of Phase 1.

Type		Value	Description
Gravity		3.73 m/s^2	Aligned with the negative y-axis
Pressure	External	0.9 kPa	Applied to the outer surface of the structures uniformly
	Internal	0.9 kPa	Applied to the internal surface of the structures uniformly
Temperature	External	$-120 \sim +20 \text{ }^\circ\text{C}$	Applied to the outer surface of the structures circularly
	Internal	$-120 \sim +20 \text{ }^\circ\text{C}$	Applied to the internal surface of the structures
Fixture		N/A	Fully constrained (fixed) at the base plate/connection to the ground

Table 6
Initial conditions for the simulation of Phase 2.

Type		Value	Description
Gravity		3.73 m/s ²	Aligned in the negative y-axis
Pressure	External	0.9 kPa	Applied to the outer surface of the regolith uniformly
	Internal	101.325 kPa	Applied to the internal surface of the structures uniformly
Temperature	External	−120~+20 °C	Applied to the outer surface of the structures circularly
	Internal	None	None
Fixture		N/A	Fully constrained (fixed) at the base plate/connection to the ground

the thermal conduction and mechanical response of the structure under Martian extreme environmental conditions, particular attention was given to the co-node setup between the structural components and the regolith. This approach enhances the continuity of temperature and stress fields at the interface, ensuring more precise thermal conduction and mechanical responses at the contact surfaces. It also improves the representation of thermal coupling effects between materials, thereby enhancing simulation reliability and stability.

Additionally, to more accurately simulate the impact of Martian temperature fluctuations, the steel regions were refined. This refinement captures the temperature gradient across the material thickness, ensuring that thermal stress analysis reflects actual material properties. In an environment with alternating high and low temperatures, slight differences in thermal expansion can lead to material fatigue and crack propagation, making mesh refinement essential for a more accurate representation of this process.

These parts were meshed using 8-node linear heat transfer brick (DC3D8) elements. The convective and radiation boundary conditions were applied in the Heat Transfer step as surface film and surface radiation conditions, respectively. The initial temperature condition of 20 °C was applied as a Predefined temperature field.

3.3. Setup

Multiple simulation scenarios were designed to compare and analyze the durability and reliability of different structural configurations under varying service periods and temperature cycles (Table 7), as well as to gain a deeper understanding of how the protective effect of regolith influences the structural fatigue life and thermal stability. Since one Martian year is equivalent to 687 Earth days, the thermal cycling count is calculated as multiples of 687.

4. Results and discussion

4.1. Temperature field analysis

4.1.1. Temperature distribution in the regolith depth

Understanding how the temperature varies through the depth of the regolith cover is crucial for the design of these structures to accurately assess the thermal stresses and deflections that can be expected within the structures (Malla and Brown, 2015). A detailed analysis of the surface temperature of the structure after 687 temperature cycles was conducted, comparing the thermal response with and without regolith protection. In Phase 1, the structure was exposed to Martian surface conditions, with its surface

Table 7
Setups.

No.	Structure Type	Phase	Condition	Thickness of Shielding Regolith Layer (m)	Thermal Cycles (Earth days)	Description	Job name
1	Arch	1	Unweathered structure	0	0	Spontaneous Crack Development	Arch-C0-P1
2		1	Unweathered structure	0	687		Arch-C687-P1
3		2	Weathered structure	3	687		Arch-C687-P2
4	Dome	1	Unweathered structure	0	0		Dome-C0-P1
5		1	Unweathered structure	0	687		Dome-C687-P1
6		2	Weathered structure	3	6870		Dome-C687n-P2
7	Cylinder	1	Unweathered structure	0	0		Cyli-C0-P1
8		1	Unweathered structure	0	687		Cyli-C687-P1
9		2	Weathered structure	3	687		Cyli-C687-P2

Basic Assumptions:

(1) The environmental temperature is assumed to be homogeneous to simplify the temperature distribution model;(2) The impact of external temperature fluctuations on structural components is the primary focus, with convective heat transfer and temperature regulation within the internal living space not considered in Phase 2;(3) The temperature at the bottom of the structure is assumed to be steady due to direct contact with the Martian surface, so temperature input at the bottom surface is not considered.

Note: C0 = 0 thermal cycles, C687 = 687 thermal cycles, P1=Phase 1, P2=Phase 2.

temperature fluctuating periodically in response to environmental temperature changes, ranging from -105.2°C to 13.5867°C . The high thermal conductivity of steel led to rapid internal-external temperature equalization (Figs. 6 and 7). In Phase 2, regolith coverage significantly mitigated temperature fluctuations. During low-temperature cycles, the surface temperature of the steel structures underwent drastic changes, creating a large temperature gradient and alternating stresses and strains.

Temperature measurements were taken at various depths within the regolith for the Dome configuration, including depths of 0 m, 0.5 m, 1 m, 1.5 m, 2 m, 2.5 m, and 3 m (Fig. 8). This configuration was selected as representative,

given the similarity in thermal response observed across all structural types. As thermal cycles increased, heat gradually transferred from the regolith surface to the interior. Regions closer to the steel structures' surface exhibited a more stable temperature variation trend (Fig. 9). The regolith surface (0 m), exposed to the environment, fluctuated between -109.275°C and $+15.973^{\circ}\text{C}$. This temperature variation occurs because the Martian surface and its thin atmosphere significantly diminish convective heat transfer between the atmosphere and the surface regolith, preventing the surface temperature from dropping as low as -120°C . While the temperatures at other points showed a gradual decrease, reaching minimum values at the end of

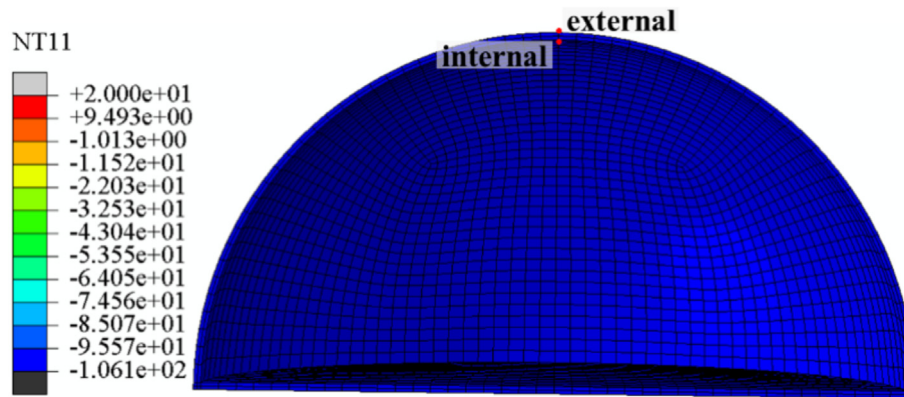


Fig. 6. Phase 1 – Temperature distribution and measuring points (Dome).

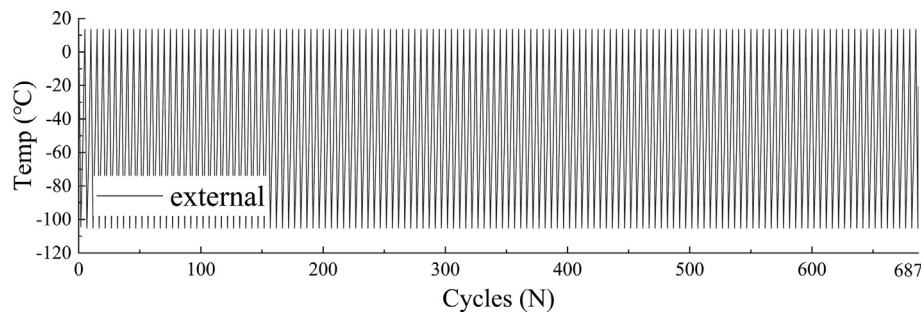


Fig. 7. Phase 1 – Temperature-Time profile on the external surface of the steel structure.

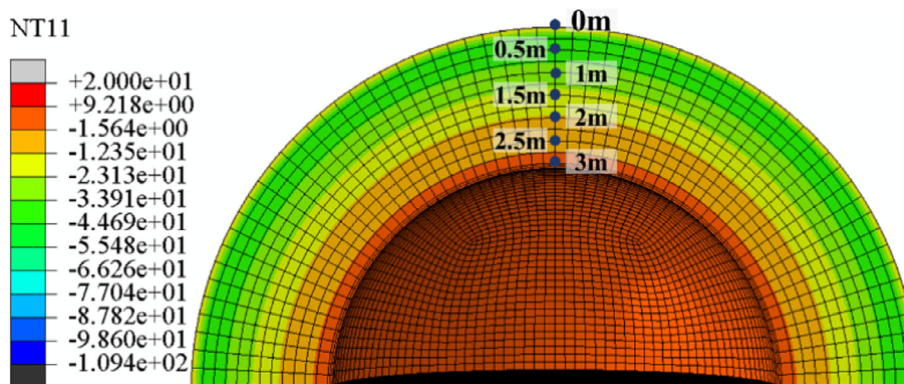


Fig. 8. Phase 2 – Temperature distribution and measuring points (Dome).

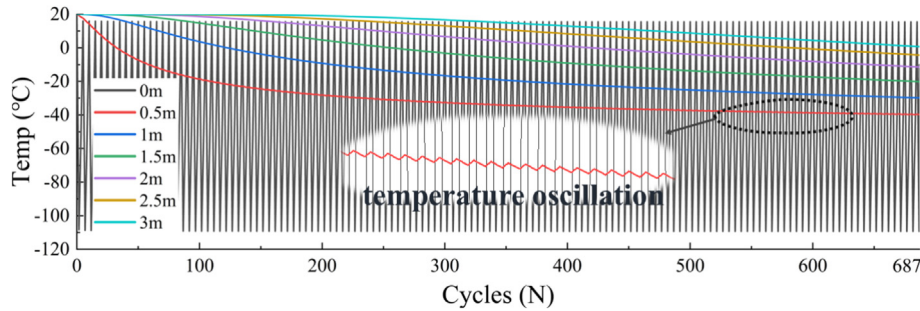


Fig. 9. Phase 2 – Temperature-Time profile along regolith depth.

Table 8

Phase 2 – Temperature distribution and gradient at different depths of the regolith (687 temperature cycles).

Measuring point (Depth from the Regolith Surface /m)	Minimum Temperature within Fluctuations (°C)	Temperature Difference from Previous Depth (°C)	Temperature Gradient (°C/m)
Ambient Temperature	–120	/	/
0	–109.275	10.725	/
0.5	–39.7888	69.4862	138.9724
1	–29.7783	10.0105	20.021
1.5	–20.1285	9.6498	19.2996
2	–11.4344	8.6941	17.3882
2.5	–4.3109	7.1235	14.247
3 (Structure Surface)	0.69229	5.00319	10.00638

Note: Temperature Gradient (G) = $\frac{\Delta T}{\Delta x}$ Where ΔT is the temperature difference, and Δx is the corresponding distance.

the thermal cycles, including -39.789 °C, -29.778 °C, -20.129 °C, -11.434 °C, -4.311 °C, and 0.692 °C, respectively. The temperature gradient decreased with depth, indicating slower temperature variation and response (Table 8). While the 0.5 m depth still experienced some fluctuation, its amplitude was significantly reduced, effectively damping the general trends of surface temperature fluctuations and demonstrating the regolith's thermal insulation effectiveness.

The results show that regolith effectively isolates the structures from external temperature extremes, significantly reducing heat transfer and the temperature reaching the steel structure surfaces, thereby mitigating the impact of Martian surface temperature fluctuations on the structures.

4.1.2. Structural temperature distribution under different temperature cycles

(1) Temperature Distribution along the Regolith Depth

A total of 6870 thermal cycles were conducted to analyze temperature variations at different depths within the regolith. The results show that as thermal cycles increase, the temperature at various depths gradually stabilizes, reaching approximately -50 °C (Fig. 10). This indicates that the regolith, due to its thermal conductivity, initially responds quickly to extreme temperature fluctuations but eventually reaches thermal equilibrium. This buffering

effect stabilizes the surface temperature of the steel components in contact with the regolith and significantly reduces the potential impact of thermal fatigue, preserving the structure's integrity.

Further analysis reveals that as thermal cycles progress, heat flux within the regolith decreases and temperature fluctuations diminish, emphasizing its role as thermal resistance. Moreover, this thermal equilibrium reduces energy consumption and protects the structure from harsh environmental conditions, supporting its long-term stability and reliability under extreme temperature fluctuations.

(2) Variation of Minimum Surface Temperature of Steel Structures

To investigate the thermal insulation effect of the regolith and demonstrate how it reduces energy consumption and enhances the thermal stability of the structure under extreme environmental conditions, the minimum surface temperature of the steel structure after 687n thermal cycles was extracted. A relationship between the minimum temperature and the number of environmental thermal cycles was established, which is expressed by Eq. (6):

$$T_{min} = 73.327 \times e^{\left(\frac{-c}{1488.945}\right)} - 50.956 \quad (6)$$

where c is the number of environmental thermal cycles, and T_{min} is the minimum surface temperature of the steel structure.

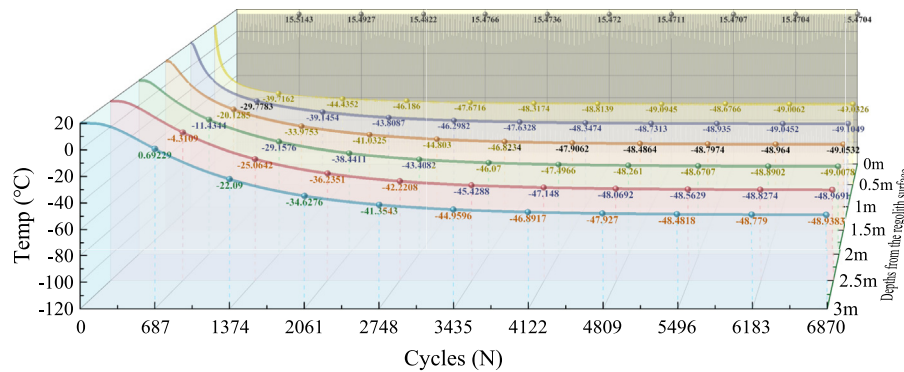


Fig. 10. Temperature-Time history along the regolith thickness direction for different thermal cycles.

This equation reflects how the structure gradually reaches thermal equilibrium under sustained thermal cycling. The parameters 73.327 and 1488.945 represent the amplitude of temperature variation and the time constant, respectively. As the thermal cycles progress, the regolith's thermal insulation effect becomes more pronounced. After an initial significant decrease, the steel structure's surface temperature gradually stabilizes (Fig. 11). This indicates that, with prolonged cycling, the regolith's buffering effect strengthens, leading to a stable surface temperature and further quantifying its role in enhancing the thermal stability and long-term reliability of the steel structure.

4.2. Impact of temperature cycles on the structures

Thermal cycling has a multifaceted impact on structural performance, including stress distribution, displacement changes, and failure modes. Stress response results can serve as a reference for evaluating the effectiveness of a structural configuration in distributing or bearing loads, thereby reducing the risk of structural failure. Displacement changes reflect the degree of structural deformation, which can be used to compare the rigidity and stability of the structures. Failure modes provide insights into predicting potential risks and design optimization. The following sections describe the response characteristics of three typical structural configurations under simulated temperature cycling (Figs. 12 and 13).

(1) Stress distribution variation

Without thermal cycling, stress concentration was observed at specific locations in each structural configuration. For the Arch, Dome, and Cylinder configurations, the maximum stresses occurred at the following locations: the sharp corners of the floor member, the junction between the floor member and the dome member, and the junction between the sidewall and the cylindrical member, with maximum stresses of 0.7116 MPa, 0.2328 MPa, and 7.227 MPa, respectively. After 687 thermal cycles, stress concentrated at the fixed bottom of the three structures, with the maximum stress reaching 205 MPa,

surpassing the material's yield strength and leading to structural failure. No significant stress changes were observed at the top (as shown in Figs. 12 and 13, panels (a), (b), and (c)). This phenomenon is primarily due to the fixed constraints at the floor member, which prevent expansion and contraction from thermal cycles, causing stress accumulation until the yield limit was reached.

(2) Displacement change analysis

Before and after thermal cycling, all configurations showed the least displacement at the fixed bottom, indicating stability at the fixed points under thermal stress. Thermal expansion and contraction caused differing stress-displacement responses across regions, particularly at connection points and areas of more flexible deformation in the upper part of the structures. For these three configurations, the maximum displacements before thermal cycling were 0.03009 mm, 0.006505 mm, and 0.07759 mm, and after thermal cycling, they were 2.693 mm, 1.665 mm, and 11.08 mm, respectively. For the Arch configuration, the maximum displacement shifted from the center of the arch apex to the sides of the arch member; for the Dome configuration, the maximum displacement expanded from the top of the dome member to the entire structure; and for the Cylinder configuration, the maximum displacement shifted from the sides of the cylindrical member to the junction between it and the upper sidewall (as shown in Figs. 12 and 13, panels (d), (e), and (f)).

(3) Structural failure modes

Without thermal cycling, none of the configurations showed significant failure. However, after introducing temperature cycling, thermal fatigue cracks appeared in all structures. In the Arch configuration, cracks occurred at the tips of the floor member; in the Dome configuration, cracks initiated at the junction between the floor and dome member, extending upward along the dome; in the Cylinder configuration, cracks appeared at the junction between the sidewall and the base of the cylindrical member, expanding toward the center of the members. Thermal expansion and contraction were restrained by the fixed

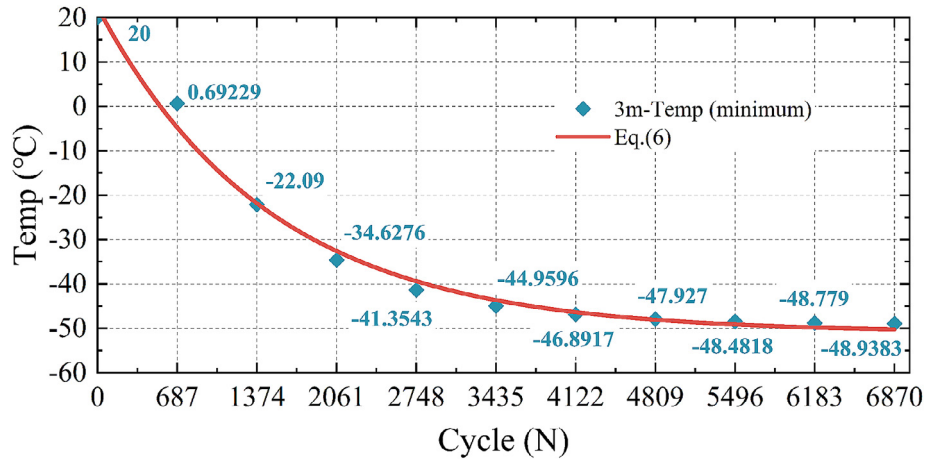


Fig. 11. Minimum surface temperature (3 m) of steel components after different thermal cycles.

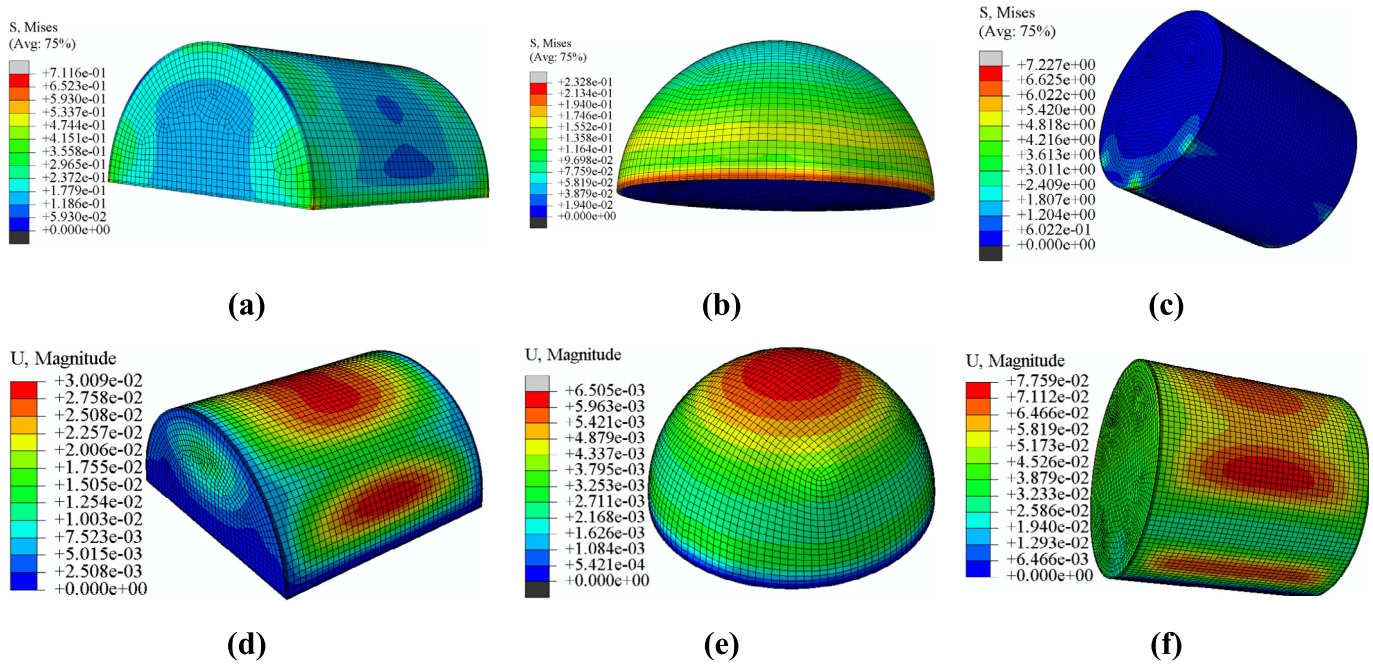


Fig. 12. Stress and displacement distribution for configurations at 0 cycle (Phase 1): (a) Arch-S; (b) Dome-S; (c) Cylinder-S; (d) Arch-U; (e) Dome-U; (f) Cylinder-U.

base plates, while upper structure displacements caused significant deformations, especially at connection points (as shown in Figs. 12 and 13, panels (g), (h), and (i)). Thus, after thermal cycling, structural failure typically occurs at connection points or areas with geometric discontinuities.

(4) Suggestions for configuration improvement

Based on the stress distribution, displacement deformation, and failure modes of the configurations, optimization suggestions are proposed. For the Arch configuration, rounded corners at the floor member and thicker connection areas are recommended to alleviate stress concentration. For the Dome configuration, improvements to the

geometric design of the base-ground connection are recommended, such as adding a base support structure or using multi-layer flexible connections. For the Cylinder configuration, thickening the sidewalls and improving the base connection design can optimize transition areas, reducing stress concentration.

4.3. Impact of temperature cycles under pressure differences

To assess the service performance of the structures during two lifecycle phases, a deeper analysis is conducted on the combined effects of thermal cycling loads and internal-external pressure differentials in Phase 2. The

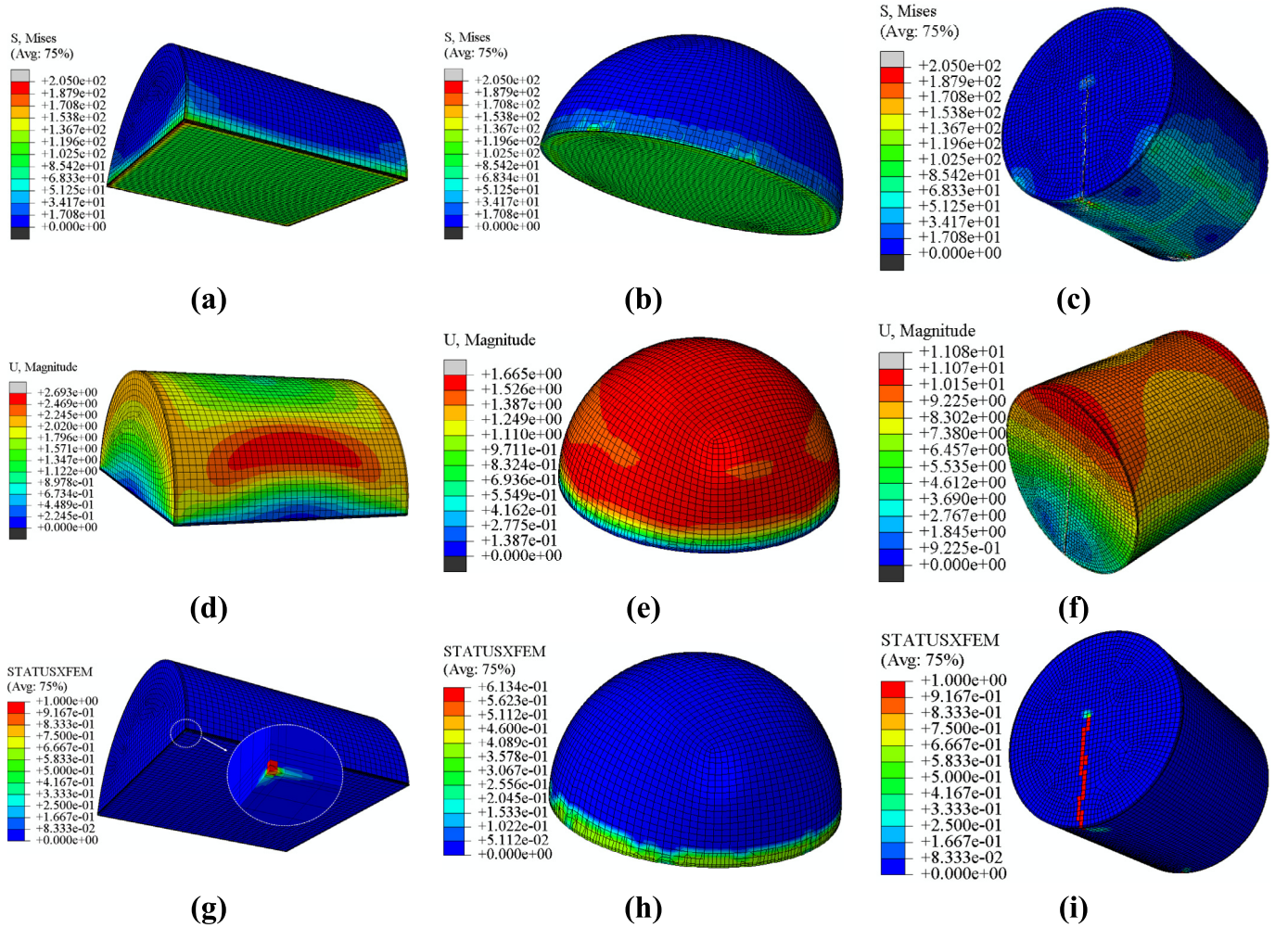


Fig. 13. Stress and displacement distribution for configurations at 687 cycles (Phase 1): (a) Arch-S; (b) Dome-S; (c) Cylinder-S; (d) Arch-U; (e) Dome-U; (f) Cylinder-U; (g) Arch-Statusxfem; (h) Dome-Statusxfem; (i) Cylinder-Statusxfem.

pressure differential significantly complicates the structure's behavior, leading to differences in stress distribution and displacement.

In Phase 1, the structures are unprotected by regolith shielding, and the effects of thermal cycling are more pronounced, resulting in higher stress and displacement. In Phase 2, the 3-meter regolith shielding reduces the steel structures' surface temperature to 0.692°C after 687 cycles, protecting them from temperature fluctuations and decreasing the temperature gradient. This significantly reduces thermal stress and deformation, as evidenced by lower maximum stress and displacement values, as seen in Figs. 13 and 14. According to Eqs. (7) and (8), the maximum stresses for the three configurations are 205 MPa, 148.4 MPa, and 10.35 MPa, showing reductions of 0 %, 27.61 %, and 94.95 %, respectively, compared to the maximum stress after 687 cycles in Phase 1. Displacement values decrease to 2.105 mm, 1.480 mm, and 2.180 mm, with reductions of 21.83 %, 11.11 %, and 80.32 %, respectively. Only the Arch configuration experiences structural failure,

with cracks appearing at the sharp corners of the floor member (Fig. 14(g)), like the situation in Phase 1 under 687 thermal cycles (Fig. 13(g)). These results demonstrate that the regolith shielding provides the most significant protective effect for the Cylinder configuration.

$$\begin{aligned}
 \text{Maximum stress reduction rate} &= \frac{\Delta \text{Stress}}{\text{Maximum stress}_{P1}} \\
 &\times 100\% \\
 &= \frac{\text{Maximum stress}_{P1} - \text{Maximum stress}_{P2}}{\text{Maximum stress}_{P1}} \times 100\% \quad (7) \\
 \text{Maximum displacement reduction rate} &= \frac{\Delta \text{Displacement}}{\text{Maximum displacement}_{P1}} \times 100\% \\
 &= \frac{\text{Maximum displacement}_{P1} - \text{Maximum displacement}_{P2}}{\text{Maximum displacement}_{P1}} \\
 &\times 100\% \quad (8)
 \end{aligned}$$

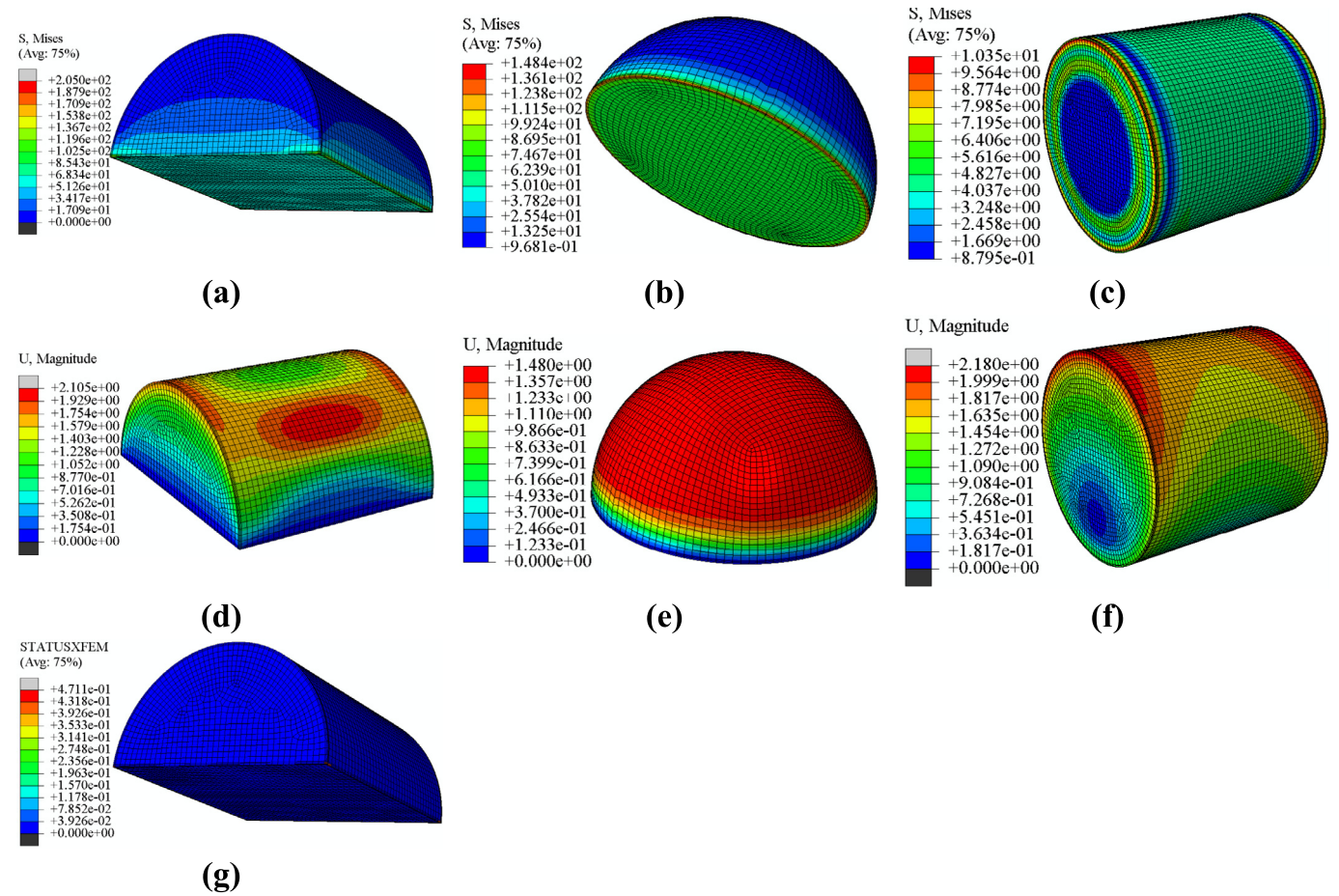


Fig. 14. Stress and displacement distribution for configurations at 687 cycles (Phase 2): (a) Arch-S; (b) Dome-S; (c) Cylinder-S; (d) Arch-U; (e) Dome-U; (f) Cylinder-U; (g) Arch-Statusxfem.

Table 9
Maximum stress, maximum displacement, and failure mode parameters of different structural configurations at different life cycle stages.

Structural Configuration	Lifecycle Phase	Thermal Cycles (Times)	Maximum Stress (MPa)	Maximum Displacement (mm)	Failure Mode
Arch	Phase 1	0	0.7116	0.03009	No Failure
		687	205	2.693	Failure occurs at the floor member (4 corners)
	Phase 2	687	205	2.105	Failure occurs at the floor member (4 corners)
Dome	Phase 1	0	0.2328	0.006505	No Failure
		687	205	1.665	Failure occurs (Cracks are uniformly distributed at the bottom of the dome member)
	Phase 2	687	148.4	1.480	No Failure
Cylinder	Phase 1	0	7.227	0.07759	No Failure
		687	205	11.08	Failure occurs at the junction of the sidewall and floor member, extending toward the center of the panel.
	Phase 2	687	10.35	2.180	No Failure

Note: In the formulas, P2 refers to Phase 2 (after 687 thermal cycles), and P1 refers to Phase 1 (after 687 thermal cycles).

Under the combined effects of regolith shielding and internal-external pressure differentials, stress and displacement distributions change. For the Cylinder configuration,

stress predominantly concentrates at the outer edges of the cylindrical and sidewall members, with lower stress in the center of the sidewall. Overall, the influence of thermal cycling loads on the structure is not governed by a single constraint; rather, the structure's geometry, the degree of constraint, and the internal-external pressure differential,

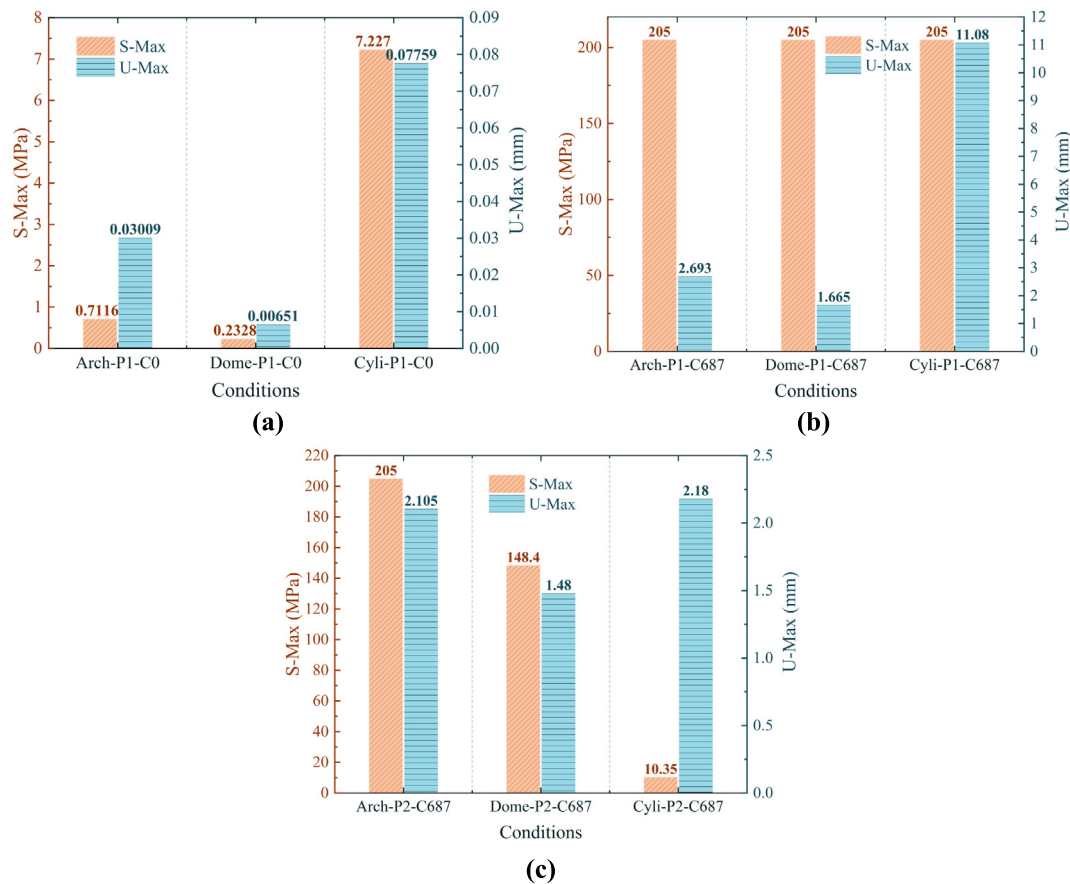


Fig. 15. Maximum stress and displacement values of three configurations at different phases under different thermal cycling counts: (a) C0-P1; (b) C687-P1; (c) C687-P2.

Table 10
Maximum stress and maximum displacement values of different structural components.

Structural Configuration	Lifecycle Phase	Maximum Stress (MPa)			Maximum Displacement (mm)		
		Vault	Base	Sidewall	Vault	Base	Sidewall
Arch	Phase 1	0.17299	0.00228	0.02044	0.02825	0	1.73e-05
		7.86169	205	83.0019	5.44532	0	5.72638
Dome	Phase 2	6.08146	57.2791	18.0294	1.05306	0	0.70776
		0.07444	0.00181	/	0.00648	0	/
Cylinder	Phase 1	0.65821	205	/	6.91095	0	/
		2.4039	67.3935	/	1.47989	0	/
	Phase 2	0.13653	/	0.28701	0.06026	/	0.0388
		2.05552	/	74.1153	9.50141	/	7.17782
	Phase 2	4.45096	/	1.17585	1.71785	/	0.59461

all of which collectively govern the final stress distribution and displacement.

4.4. Evaluation of structural configuration performance

An analysis of the maximum stress, maximum displacement, and failure mode parameters for the three structural configurations across different lifecycle phases is provided (Table 9). After 687 thermal cycles, all structures exhibit relatively small displacements, indicating good stiffness.

A comparison of maximum stress and displacement values before and after thermal cycling in Phase 1 shows that the Dome configuration exhibits the lowest stress and displacement, reflecting superior stiffness and structural stability. It exhibits the greatest resistance to thermal fluctuations, making it the most robust configuration among the three (Fig. 15).

Regarding failure modes, the Dome configuration fails uniformly at the bottom of the dome member. Cracks are evenly distributed, preventing sudden catastrophic failure

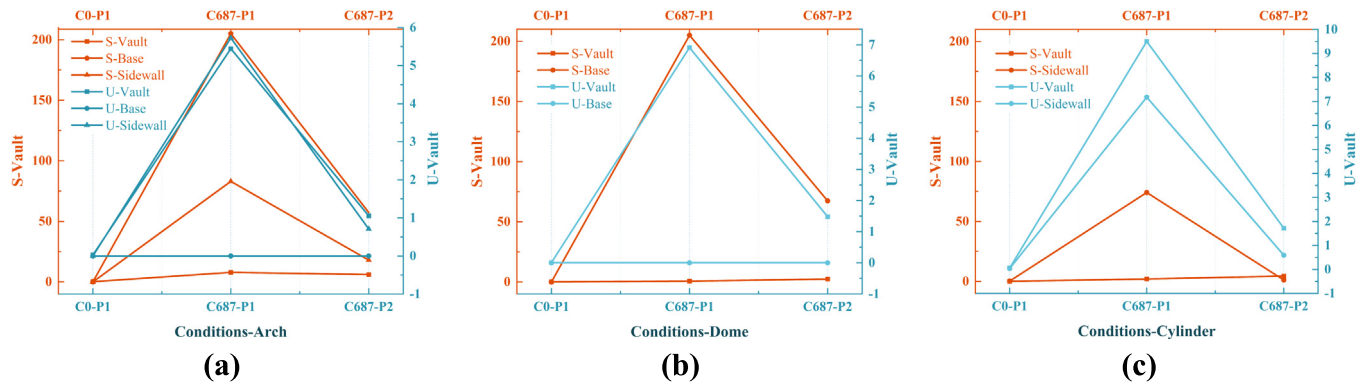


Fig. 16. Maximum stress and displacement at the centers of different components: (a) Arch; (b) Dome; (c) Cylinder.

and dispersing forces from external loads or internal pressure variations, thus reducing failure risk in practical applications. In contrast, the Arch and Cylinder configurations exhibit stress concentrations, revealing potential vulnerable areas in the structural design that require further optimization, such as thickening the sidewall components or reducing sharp corners, to improve overall stability and resistance to failure. Thus, the Dome configuration demonstrates superior performance in terms of stress distribution, stability, and resistance to sudden failure, making it more suitable for the Martian extreme environmental conditions.

A comparative analysis was performed on the maximum values observed at the center of key components across the different configurations (Table 10, Fig. 16). Horizontal comparison (different parts) reveals that under thermal cycling conditions (C687-P1/P2), the maximum stress occurs at the fixed position of floor members, where expansion and contraction are restricted, initiating failure, which requires particular attention. The vault, in comparison to the other two parts, exhibits lower stress, indicating that its curvature helps disperse the stress caused by thermal expansion. In contrast, the sidewall exhibits higher stress than the vault, suggesting that attention should be given to thickening the sidewall components. Longitudinal comparison (different phases) shows that the construction phase (C687-P1) is the most vulnerable stage for structural failure. To reduce thermal stress damage, techniques like optimizing structural weaknesses and applying pre-stressing are recommended.

Through a comparative analysis of several configurations, the Arch configuration demonstrates high structural efficiency and cost-effectiveness, as it can effectively bear gravity and transfer vertical loads to support points via the arch shape, reducing material usage. It is suitable for resource-limited scenarios, as it achieves high load-bearing capacity with less material. However, its resistance to lateral and uneven loads is weaker, making it prone to instability when support conditions change. The Dome configuration provides uniform load distribution, high spatial utilization, and natural stability. It offers the greatest volume for a given surface area, has uninterrupted intermediate space, and evenly distributes loads, making it suitable

for scenarios where pressure is uniformly applied to the top. NASA and ESA are exploring dome-shaped habitat structures (Tripathi et al., 2023). Nevertheless, cracks propagating upward in Dome configurations suggest that the curved surface is more susceptible to deformation under thermal cycling. The Cylinder configuration, with its vertical sidewalls and curved top, facilitates vertical load transfer, resists lateral pressure, and is easily modular and scalable. Compared to spheres, cylinders offer more efficient use of space (Jablonski and Ogden, 2008). Yet, the connection between the sidewalls and bottom is vulnerable to thermal cycling, as stress concentration is more likely at the transition from curved to flat surfaces due to uneven thermal expansion.

5. Conclusions

Finite element models of three basic extraterrestrial structural configurations were established to analyze the effects of Martian low-temperature thermal cycling loads on structural fatigue, covering the structural response across different lifecycle phases. The study focused on stress concentration, displacement deformation, and fatigue failure modes. The main conclusions are as follows:

- (1) A formula correlating the minimum surface temperature of the steel structure under regolith protection to the number of thermal cycles was derived, reflecting the process of the structure reaching thermal equilibrium. After 6870 thermal cycles, the surface temperature stabilized around -50°C , confirming the 3-meter regolith shielding's effective role in mitigating temperature fluctuations.
- (2) Comparative analysis showed that thermal cycling and internal-external pressure differences significantly alter stress distribution and displacement in the three configurations. Without regolith protection, cycling caused damage and crack propagation. In contrast, under regolith protection, the maximum stress reduction rates for the three configurations were 0 %, 27.61 %, and 94.95 %, while the displacement reduction rates were 21.83 %, 11.11 %, and 80.32 %, respectively.

tively. The 3-meter regolith shielding provided the most significant protection to the Cylinder configuration.

- (3) The Dome configuration performed best in terms of stress dispersion, preventing sudden failure, and maintaining stability, making it the most suitable choice under Martian extreme conditions. Optimization strategies were proposed: strengthen the Arch configuration's floor member connections with rounded transitions to reduce stress concentration; reinforce the connection between the floor and dome members for the Dome configuration; and increase the Cylinder configuration's sidewall thickness and optimize the bottom transition to reduce stress concentration.

These findings offer key insights for selecting and optimizing Martian habitat designs, propose viable strategies, and provide a framework for evaluating durability in extreme environments. It is anticipated that these results will provide valuable technical insights and contribute to the development of design standards for future Martian habitat construction. Subsequent research will develop a comprehensive evaluation system incorporating weighted multi-criteria scoring — including maximum thermal stress, displacement deformation, and fatigue life — to enable precise adaptability comparisons among structures facing cryogenic Martian conditions. Future studies should also integrate cryogenic thermal cycling experiments to advance understanding of microstructural damage evolution mechanisms, thereby refining model parameter boundaries. Collectively, this work provides theoretical underpinnings for transitioning SpaceX Starship material systems to Martian construction applications.

Data availability statement

All data, models, and code generated or used during the study appear in the submitted article.

Acknowledgement

This study was partially supported by the National Natural Science Foundation of China (Grant No. 52378216), the Chongqing Natural Science Foundation for Outstanding Young Scientists Fund (Grant No. CSTB2024NSCQ-JQX0027), and the Support Plan for Returned Overseas Scholars of Chongqing (cx2024004).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- Benaroya, H., 2017. Lunar habitats: a brief overview of issues and concepts. *Reach* 7, 14–33. <https://doi.org/10.1016/j.reach.2018.08.002>.
- Bhattacharya, B., Ellingwood, B., 1998. Continuum damage mechanics analysis of fatigue crack initiation. *Int. J. Fatigue* 20 (9), 631–639.
- Bucklin, R.A., Fowler, P.A., Rygalov, V.Y., Wheeler, R.M., Mu, Y., Hublitz, I., Wilkerson, E.G., Bucklin, R.A., 2004. Greenhouse design for the Mars environment: Development of a prototype, deployable dome. In: VII International Symposium on Protected Cultivation in Mild Winter Climates: Production, Pest Management and Global Competition 659, pp. 127–134.
- Callister Jr, W.D., Rethwisch, D.G., 2020. *Materials science and engineering: an introduction*. John Wiley & sons.
- Emmel, M.G., Abadie, M.O., Mendes, N., 2020. New external convective heat transfer coefficient correlations for isolated low-rise buildings. *Energ. Buildings* 39 (3), 335–342. <https://doi.org/10.1016/j.enbuild.2006.08.001>.
- <https://mars.nasa.gov/insight/mission/weather/>.
- <https://starwalk.space/zh-Hans/news/mars-the-ultimate-guide>.
- Incropera, F.P., DeWitt, D.P., Bergman, T.L., Lavine, A.S., 1996. *Fundamentals of Heat and Mass Transfer* Vol. 6, 116.
- Jablonski, A.M., Ogden, K.A., 2008. Technical requirements for lunar structures. *J. Aerosp. Eng.* 21 (2), 72–90. [https://doi.org/10.1061/\(ASCE\)0893-1321\(2008\)21:2\(72\)](https://doi.org/10.1061/(ASCE)0893-1321(2008)21:2(72)).
- Jarvstrat, N., Toklu, C., 2004. Design and Construction for Self-sufficiency in a Lunar Colony. ASEM2004, URL: www.moonisru.com/information/ASEM2004.pdf. https://www.researchgate.net/publication/228918710_Design_and_Construction_for_Self-sufficiency_in_a_Lunar_Colony.
- Katzer, J., Suchocki, C., Błaszczak-Bak, W., Zarzycki, P.K., Damięcka-Suchocka, M., 2021. Reliability and effectiveness of laser scanners in future construction efforts on the Moon and Mars. *Autom. Constr.* 132, 103979. <https://doi.org/10.1016/j.autcon.2021.103979>.
- Kim, C.S., 1975. Thermophysical properties of stainless steels (No. ANL-75-55). Argonne National Lab., Ill. (USA).
- Kuster, L.E., Delorit, J.D., 2023. Viability of Construction Material within an Extraterrestrial Environment. In: *In Earth and Space 2022*, pp. 602–615.
- Liu, Yufeng, Xia, Chunjing, Yan, Ming, et al., 2008. Simulation of thermal fatigue crack opening process by finite element method. *Fail. Anal. Prevent.* 3 (1), 43–47.
- Malla, R.B., and Brown, K.M., 2012. Frequency Analysis of a Frame-Membrane Lunar Habitat Considering Extreme Temperatures and Internal Pressure. In: *Earth and Space 2012: Engineering, Science, Construction, and Operations in Challenging Environments*. pp. 1400–1412. <https://doi.org/10.1061/9780784412190.149>.
- Malla, R.B., Brown, K.M., 2015. Determination of temperature variation on lunar surface and subsurface for habitat analysis and design. *Acta Astronaut.* 107, 196–207. <https://doi.org/10.1016/j.actaastro.2014.10.038>.
- Malla, R.B., Adib-Jahromi, H.R., Accorsi, M.L., 1995. Simplified design method for braced double-skinned structure in lunar application. *J. Aerosp. Eng.* 8 (4), 189–195. [https://doi.org/10.1061/\(ASCE\)0893-1321\(1995\)8:4\(189\)](https://doi.org/10.1061/(ASCE)0893-1321(1995)8:4(189)).
- Malla, R., Brown, K., Filburn, T., 2011. Structural Design of Frame-Membrane Habitat Concepts Considering Life Support Requirements for Lunar Colonization. In: *41st International Conference on Environmental Systems*, p. 5114. <https://doi.org/10.2514/6.2011-5114>.
- Mottaghi, S., Benaroya, H., 2015. Design of a lunar surface structure. I: design configuration and thermal analysis. *J. Aerosp. Eng.* 28 (1), 04014052. [https://doi.org/10.1061/\(ASCE\)AS.1943-5525.0000382](https://doi.org/10.1061/(ASCE)AS.1943-5525.0000382).
- NASA (National Aeronautics and Space Administration), 2015. *The fact and fiction of Martian dust storms*. NASA, Washington, DC.
- Naser, M.Z., 2019. Extraterrestrial construction materials. *Prog. Mater. Sci.* 105, 100577. <https://doi.org/10.1016/j.pmatsci.2019.100577>.

- National Aeronautics and Space Administration (NASA). 2004. Mars fact sheet. Accessed online 22 January 2005: <http://nssdc.gsfc.nasa.gov/planetary/factsheet/marsfact.html>.
- Peters, G.H., Abbey, W., Bearman, G.H., Mungas, G.S., Smith, J.A., Anderson, R.C., Douglas, S., Beegle, L.W., 2008. Mojave Mars simulant—characterization of a new geologic Mars analog. *Icarus* 197 (2), 470–479. <https://doi.org/10.1016/j.icarus.2008.05.004>.
- Petrov, G.I., 2004. A permanent settlement on Mars: the first cut in the land of a new frontier Doctoral dissertation. Massachusetts Institute of Technology.
- Petrov, G., Ochsendorf, J., 2005. Building on mars. *Civ. Eng. Mag. Arch.* 75 (10), 46–53. <https://doi.org/10.1061/ciegag.0000039>.
- Ruess, F., Schaenzlin, J., Benaroya, H., 2006. Structural design of a lunar habitat. *J. Aerosp. Eng.* 19 (3), 133–157. [https://doi.org/10.1061/\(ASCE\)0893-1321\(2006\)19:3\(133\)](https://doi.org/10.1061/(ASCE)0893-1321(2006)19:3(133)).
- Scheer, F.A., Wright Jr, K.P., Kronauer, R.E., Czeisler, C.A., 2007. Plasticity of the intrinsic period of the human circadian timing system. *PLoS One* 2 (8), e721.
- Sherwood, B., Touns, L., 1992. Technical issues for lunar base structures. *J. Aerosp. Eng.* 5 (2), 175–186. [https://doi.org/10.1061/\(ASCE\)0893-1321\(1992\)5:2\(175\)](https://doi.org/10.1061/(ASCE)0893-1321(1992)5:2(175)).
- Soureshjani, O.K., Massumi, A., Nouri, G., 2023. Martian buildings: Design loading. *Adv. Space Res.* 71 (5), 2186–2205. <https://doi.org/10.1016/j.asr.2022.10.066>.
- Spohn, T., Müller, N., Knollenberg, J., Grott, M., Golombek, M.P., Plesa, A.C., et al., 2024. Mars soil temperature and thermal properties from InSight HP 3 HP³ data. *Geophys. Res. Lett.* 51 (18), e2024GL108600. <https://doi.org/10.1029/2024GL108600>.
- Task Committee on Lunar Base Structures, 1992. Overview of existing lunar base structural concepts. *J. Aerosp. Eng.* 5 (2), 159–174. [https://doi.org/10.1061/\(ASCE\)0893-1321\(1992\)5:2\(159\)](https://doi.org/10.1061/(ASCE)0893-1321(1992)5:2(159)).
- Toklu, Y.C. 2005. Optimum lunar and Martian structures and their construction. In: *Proceedings of 2nd International Conference on Recent Advances in Space Technologies*, 2005. RAST 2005. IEEE. pp. 260–264. <https://doi.org/10.1109/RAST.2005.1512573>.
- Tripathi, S., Steiner, J.T., Malla, R.B., 2023. Three-dimensional temperature profile in a dome-shaped habitat structure on the moon. *Acta Astronaut.* 204, 263–280. <https://doi.org/10.1016/j.actaastro.2022.12.040>.
- Williams, D.R. 2004. Mars fact sheet. <http://nssdc.gsfc.nasa.gov/planetary/factsheet/marsfact.html>.
- Zhang, R., Zhou, S., Li, F., 2022. Preparation of geopolymer based on lunar regolith simulant at in-situ lunar temperature and its durability under lunar high and cryogenic temperature. *Constr. Build. Mater.* 318, 126033. <https://doi.org/10.1016/j.conbuildmat.2021.126033>.